PolyutilGH args

The general syntax is:

Java -jar PolyUtilGH Reader filter1 filter2 … filename/none

In general, the program reads an input file, and propagates either numbers or polyforms through a series of filters. For example:

StdinToRep RepToForm:form=Polyomino UnorderedCounter CommaStdout free1-14.txt

StdinToRep - read an input file in OEIS b-file format.

RepToForm:form=Polyomino – use each representation (see A246521) to generate a polyomino.

UnorderedCounter – count the input polyominoes according to size.

CommaStdout – write the counts of polyominoes (in OEIS data field format) to standard output (in this case, it will look like A000105).

free1-14.txt – file containing A246521 representations of polyominoes through to size 14.

More filters will be made available over time.

Args[0] is a reader.

Reader	**Function**
stdinToRep[:nolineno=true]	Reads file named as last argument (in OEIS b-file format) and writes representations to the next filter. Use “nolineno=true” if the input file has no line numbers.

Filters

A filter may be specified just with its name, or followed by a colon and a set of parameters. No spaces are allowed.

Example: Only:from=8,to=10

Filter	**Input**	**Output to next filter**	**Function**
BiTrisectable:num=n	Polyominoes/ Polycubes	Numbers	Build all possible polyforms of n copies of the input polyforms. There may be duplicates in the output.
CommaStdout	Numbers	-	Writes comma-separated numbers to the process stdout, ready for OEIS
Condition			See specific table
EliminateDuplicates	Numbers	Numbers	Does not require sorted data; does build table in RAM
EliminateOrderedDuplicates	Numbers	Numbers	Requires (but does not verify) sorted data; does not build table
EveryInterval:interval=n	Numbers or Polyforms	Numbers or Polyforms	Will pass only every n lines; example: if n=4, it will pass lines 0,4,8,12,…
Graphics:[prefix=pppp, horiz=n, lattice=true, centre=true, haxis=true, vaxis=true, swneaxis=true, senwaxis=true, applycb=true, border=true, caption=true, round=true, markers=true, rows=m, connectedness=true, suppressinnerborders=true, reverse=true, showsym=true, connectedness=true, hmargin=nnn, vmargin=nnn,edge=nnn, centrex=float, centrey=float, diameter=float, space=n, bw=true]	Polyforms	Polyforms	Builds png images of the input polyforms; groups them according to size. Uses pppp as png file name prefix. Draws maximum n polyforms horizontally. Changes page at m rows, default 8. Behaviour depends on various flags, which are relevant to specific polyforms. **Requires existence of "png" folder.**
Only:[from=m,]to=n, perimeter=true, nostop=true	Polyforms	Polyforms	Will pass only polyforms of size >= m (default 0) and less than n; it assumes input is ordered to size and will cause filter termination once a polyform of size > n is seen (unless nostop is set) If perimeter=true, then passes only polyforms with consistent

			perimeters; does not assume ordering and does not terminate.
RepToForm:form=Polyomino/ Polycube	Numbers	Polyform	Converts a representation into the corresponding polyform, according to its name
Sort	Numbers	Numbers	Sorts the data. Side effect: mixes polyominoes of different sizes, if present. Eliminates duplicates. Builds table in RAM.
SortForm	Polyforms	Numbers	Sorts representations primarily over size, then over numeric value.
Stdout[:nolineno=true]	Polyforms or numbers representing polyforms	-	Writes the representations to the process stdout, usually with line numbers.
UnorderedCounter	Polyforms	Numbers	Will count polyforms of the same size, even if not ordered to size in input; as a result, it outputs numbers only when all data has been read

Condition filters

The condition filter passes only polyforms that satisfy a particular condition. Different conditions may apply to polyforms of different forms, as shown in the table below. Boolean conditions, XXX=true or XXX=false apply both ways; i.e. if the condition expressed is XXX=true, then only polyforms satisfying the condition XXX will be passed on to the next filter; if the condition is XXX=false, then only polyforms NOT satisfying the condition XXX will be passed on.

If more than one condition filter is included in the pipeline, they must obviously all be satisfied.

Syntax: Condition:type=value

Example: Condition:wellconnected=false

Condition type	**Values**	**Polyforms**	**Condition applied**
almostsquare	true/false	Polyominoes	Passes only polyominoes having the same width and height, or at most with a difference of 1
board[,both=true]	true/false	Polyominoes	Passes only board polyominoes; if also both=true passes only board polyominoes that are board when rotated 90 degrees. These correspond to convex polyominoes
convex	true/false	Polyominoes	Passes only convex polyominoes
convextreelike	true/false	Polyominoes	Passes only convex treelike polyominoes
evenwidth	true/false	Polyominoes	Passes only polyominoes of even width
intact	true/false	Polyominoes	Passes only polyominoes that are intact, i.e., connected.
loop	true/false	Polyominoes	Passes only loop (ring) polyforms. Questionably does not include 2x2 polyomino
loopborder	true/false	Polyominoes	Passes only polyominoes whose outer borders create a loop
maxbiased	true/false	Polyominoes	Passes only maximally-biased polyominoes
minperimeter	true/false	Polyominoes	Passes only polyominoes of minimal perimeter
minvertices	true/false	Polyominoes	Passes only polyominoes having the minimal number of vertices
perimeter	n	Polyominoes	Passes only polyominoes having perimeter n
pile	true/false	Polyominoes	Passes only pile polyominoes
selfavoiding	true/false	Polyominoes	Passes only polyominoes with a self-avoiding border; equivalent to passing polyominoes without holes
simplyconnected	true/false	Polyominoes	Passes only polyominoes whose interiors are simply connected (no “real” holes)

smallholes	true/false	Polyominoes	Passes only polyominoes whose holes, if any, have each the size of a single cell.
smooth	true/false	Polyominoes	Passes only polyominoes that have no squares connected on only one edge
square	true/false	Polyominoes	Width = height
staircase	true/false	Polyominoes	Passes only staircase polyominoes
strip	true/false	Polyominoes	Passes only strip (snake) polyforms
sym	Symmetry type list	Polyominoes	Passes only polyominoes satisfying the list of required symmetries; see below
treelike	true/false	Polyominoes	Passes only treelike polyominoes
unbiased	true/false	Polyominoes	Passes only unbiased polyominoes
wellconnected	true/false	Polyominoes	Passes only well-connected polyominoes

Condition type “sym”

The sym condition tests the input polyomino for conformance to a list of symmetries. Each symmetry may be tested positively (sym=+S – only polyominoes having symmetry S will pass) or negatively (sym=-S – only polyominoes NOT having symmetry S will pass). Multiple symmetries may be expressed; sym=+S+T will pass polyominoes satisfying either S or T symmetries. Remember that to pass polyominoes that satisfy BOTH symmetries, multiple condition filters may be invoked.

Acronym	**Symmetry**
M90	Horizontal or vertical
M90A	Horizontal or vertical, aligned to an edge
M45	Diagonal
MH	Horizontal
MV	Vertical
MSENW	\
MSWNE	/
R90	90 degree rotational
R180	180 degree rotational
R90V	90 degree rotational about a vertex
R180C	180 degree rotational about the centre of a square
R180ME	180 degree rotational about the middle of an edge